

Exp No:5 Data Collection and Initial Exploration

Objective: To collect, load, and perform initial exploration of the diabetes dataset.

Materials Needed:

- Diabetes dataset (available from sources such as UCI Machine Learning Repository or Kaggle)
- Python with libraries: pandas, numpy, matplotlib, seaborn

Procedure and Code:

1. Data Collection:

```
import pandas as pd

# Load the dataset
diabetes_df = pd.read_csv("diabetes.csv")

# Display the first few rows
print(diabetes_df.head())
```

```
[1] import pandas as pd
db= pd.read_csv("/content/drive/MyDrive/PIMA Indian Dataset.csv")
```

```
print(db.head())
```

	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	\
0	6	148	72	35	0	33.6	
1	1	85	66	29	0	26.6	
2	8	183	64	0	0	23.3	
3	1	89	66	23	94	28.1	
4	0	137	40	35	168	43.1	

	DiabetesPedigreeFunction	Age	Outcome
0	0.627	50	1
1	0.351	31	0
2	0.672	32	1
3	0.167	21	0
4	2.288	33	1

2. Initial Exploration:

```

# Dataset structure and summary statistics
print(db.info())
print(db.describe())

# Visualize the distribution of each feature
import matplotlib.pyplot as plt
import seaborn as sns

db.hist(bins=50, figsize=(20, 15))
plt.show()

# Pairplot to visualize relationships
sns.pairplot(db)
plt.show()

```

[3]

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 768 entries, 0 to 767
Data columns (total 9 columns):
#   Column                Non-Null Count  Dtype
---  ---                ---
0   Pregnancies            768 non-null   int64
1   Glucose                768 non-null   int64
2   BloodPressure          768 non-null   int64
3   SkinThickness          768 non-null   int64
4   Insulin                768 non-null   int64
5   BMI                   768 non-null   float64
6   DiabetesPedigreeFunction 768 non-null   float64
7   Age                   768 non-null   int64
8   Outcome                768 non-null   int64
dtypes: float64(2), int64(7)
memory usage: 54.1 KB
None

```

	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin
count	768.000000	768.000000	768.000000	768.000000	768.000000
mean	3.845052	120.894531	69.105469	20.536458	79.799479
std	3.369578	31.972618	19.355807	15.952218	115.244002
min	0.000000	0.000000	0.000000	0.000000	0.000000
25%	1.000000	99.000000	62.000000	0.000000	0.000000
50%	3.000000	117.000000	72.000000	23.000000	30.500000
75%	6.000000	140.250000	80.000000	32.000000	127.250000
max	17.000000	199.000000	122.000000	99.000000	845.000000

	BMI	DiabetesPedigreeFunction	Age	Outcome
count	768.000000	768.000000	768.000000	768.000000
mean	31.601570	0.471876	33.260885	0.348558
std	7.384108	0.331329	11.760232	0.476951
min	0.000000	0.070000	21.000000	0.000000
25%	27.300000	0.243750	24.000000	0.000000
50%	32.000000	0.372500	29.000000	0.000000
75%	36.000000	0.526250	41.000000	1.000000
max	67.100000	2.420000	81.000000	1.000000



